

Padé differencing + VSI results

<https://arxiv.org/abs/2406.15611>

<https://arxiv.org/abs/2411.03467>

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Padé/compact schemes are finite-difference schemes that have higher resolving power compared to non-compact schemes of the same order.

The simplest fourth-order scheme for the first-derivative is:

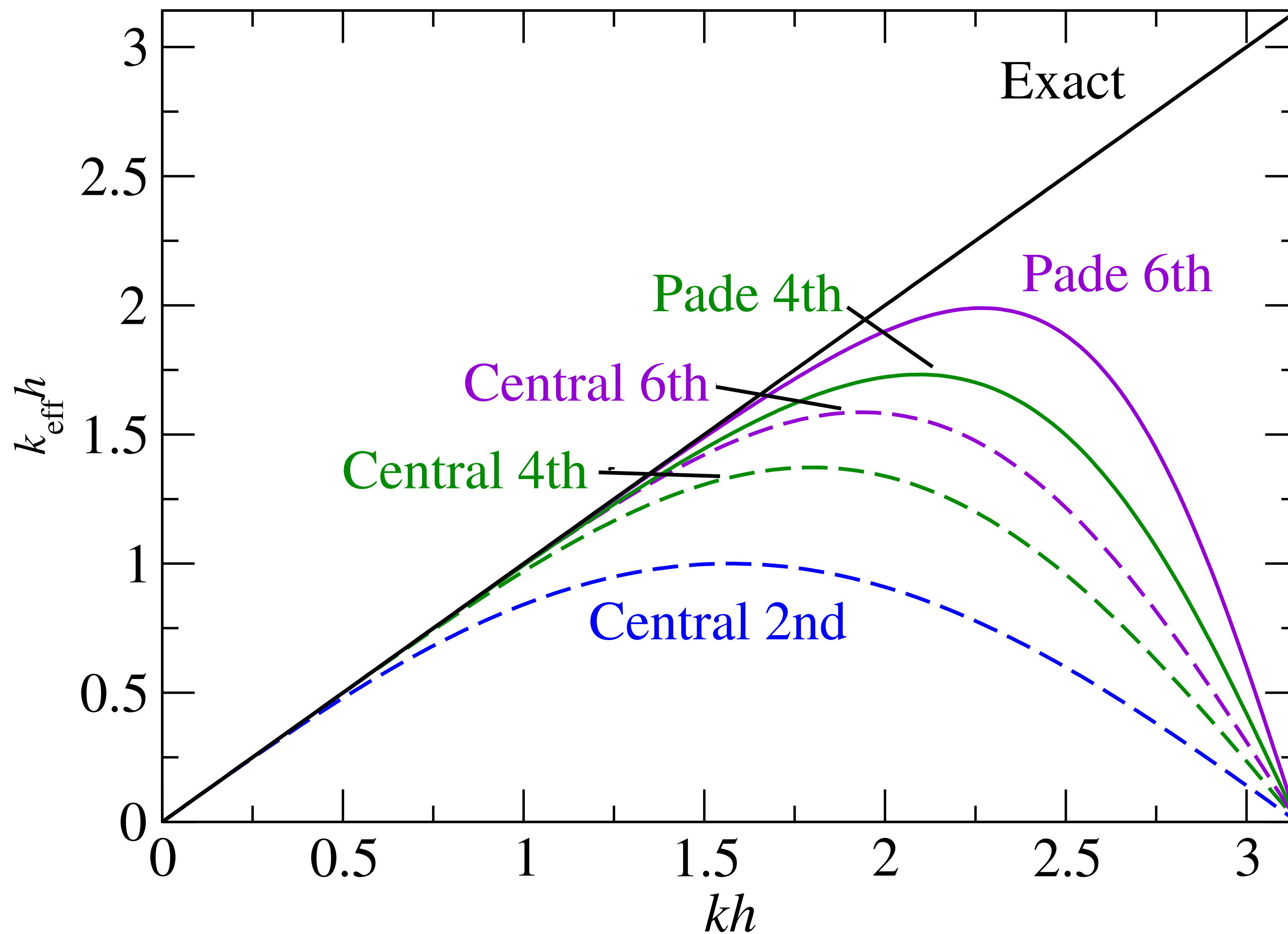
$$\alpha f'_{j-1} + f'_j + \alpha f'_{j+1} = \frac{a}{2h} (f_{j+1} - f_{j-1}) \quad \alpha = 1/4, \quad a = 3/2$$

Requires a tridiagonal solve along each line of data.

Resolving power: How well does the scheme reproduce the exact eigenvalues and eigenvectors of differential operators. For periodic BC this becomes the so-called modified wavenumber analysis.

For periodic BC, substitute above $f_j = e^{ikx_j}$. Get $f'_j = ik_{\text{eff}} e^{ikx_j}$.

(a) Effective wavenumber for numerical differences.

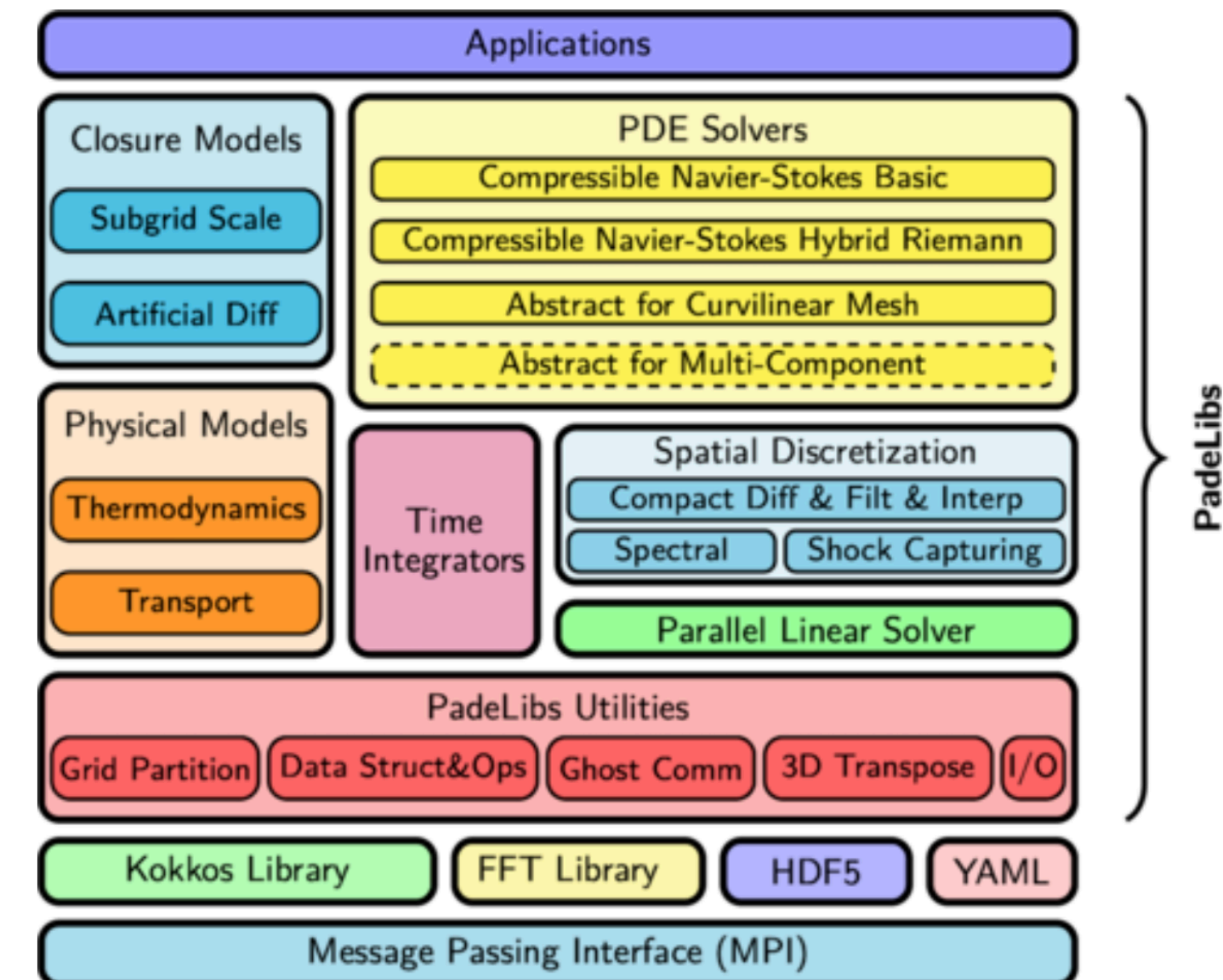
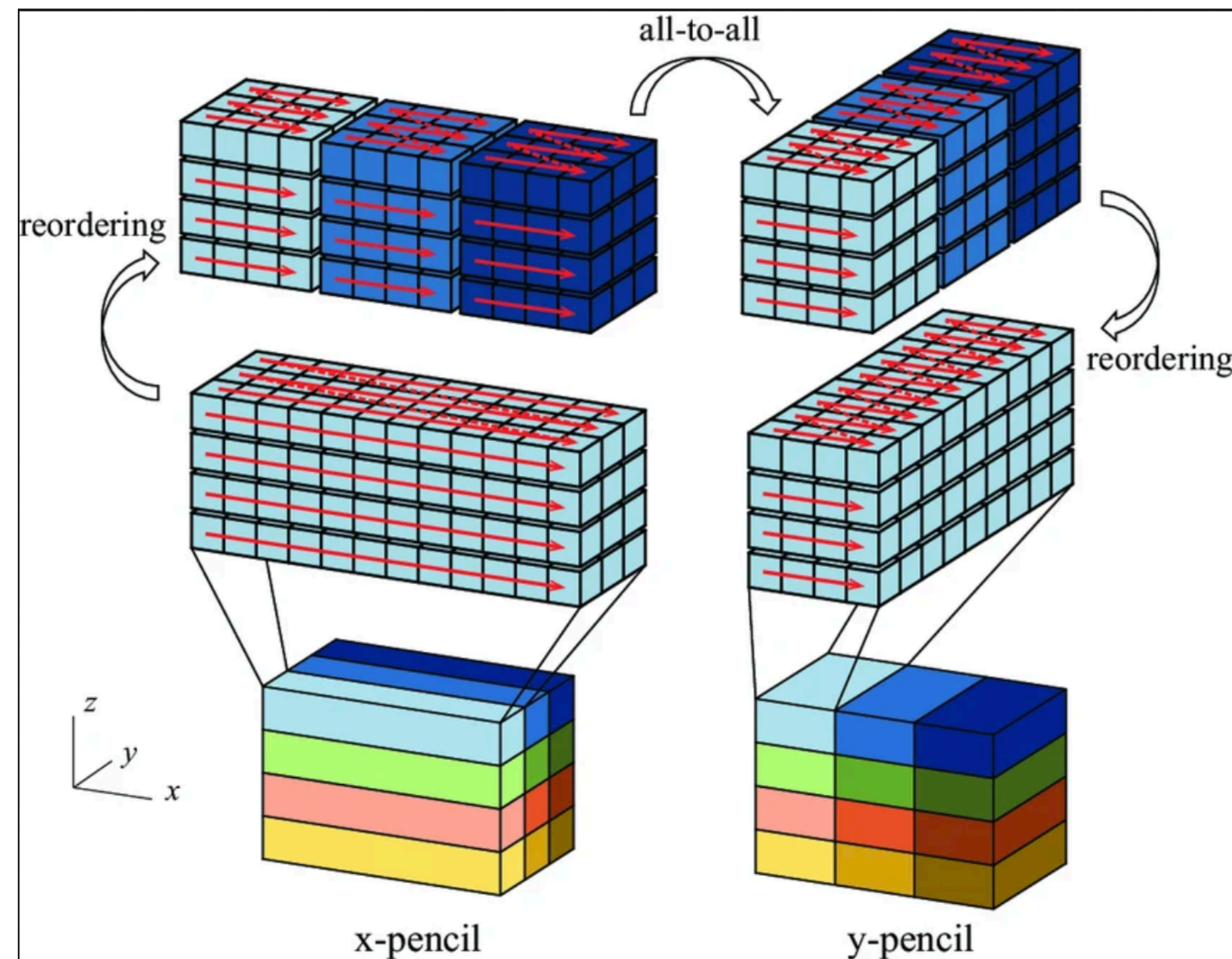


Parallel Implementation

Current Code: Pencil data structure with data transposes: communication intensive

Parallel tridiagonal solvers not requiring transposes

PadeLibs e.g., from Stanford supports both CPUs and GPUs

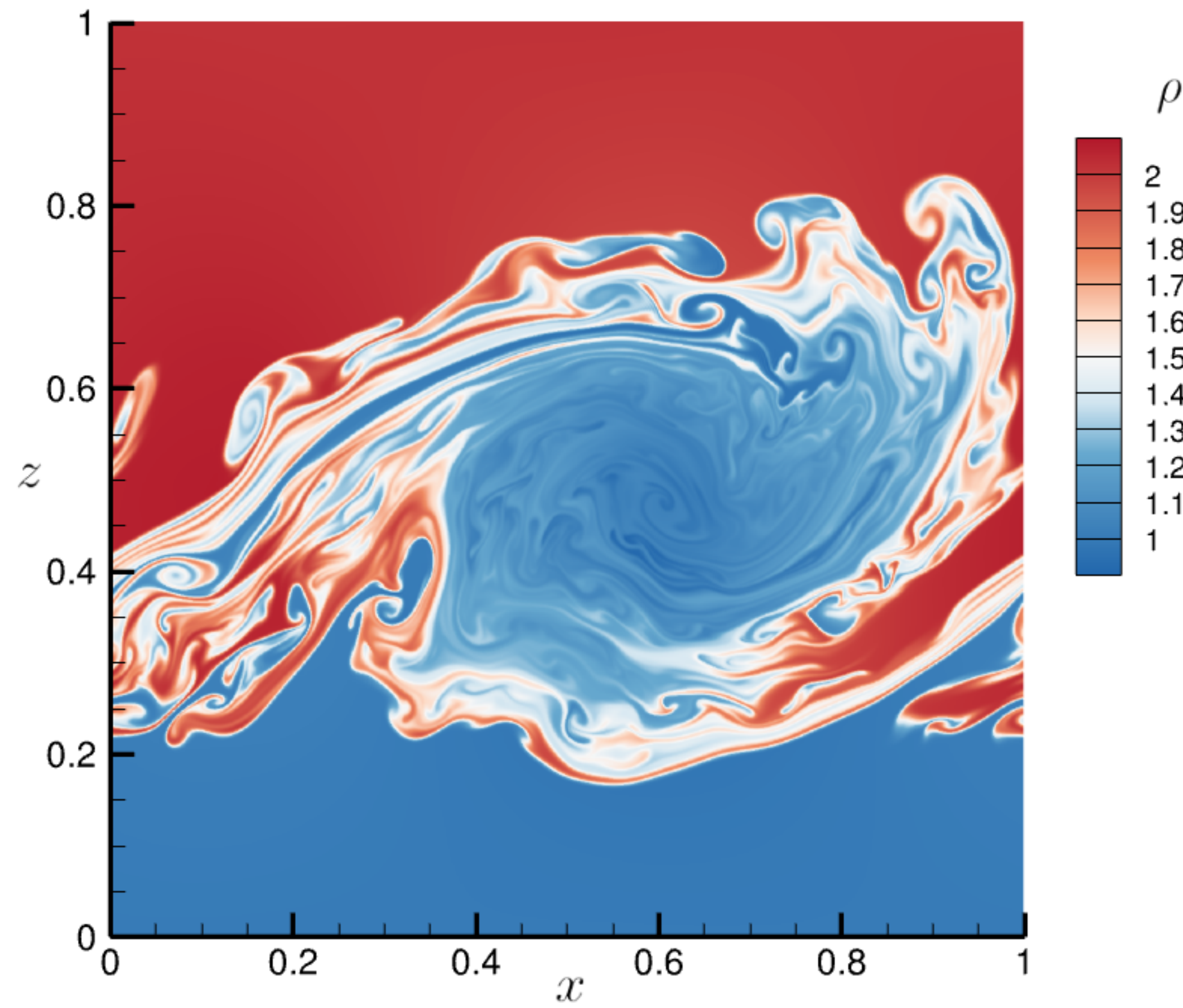


Contact: Hang Song at Stanford

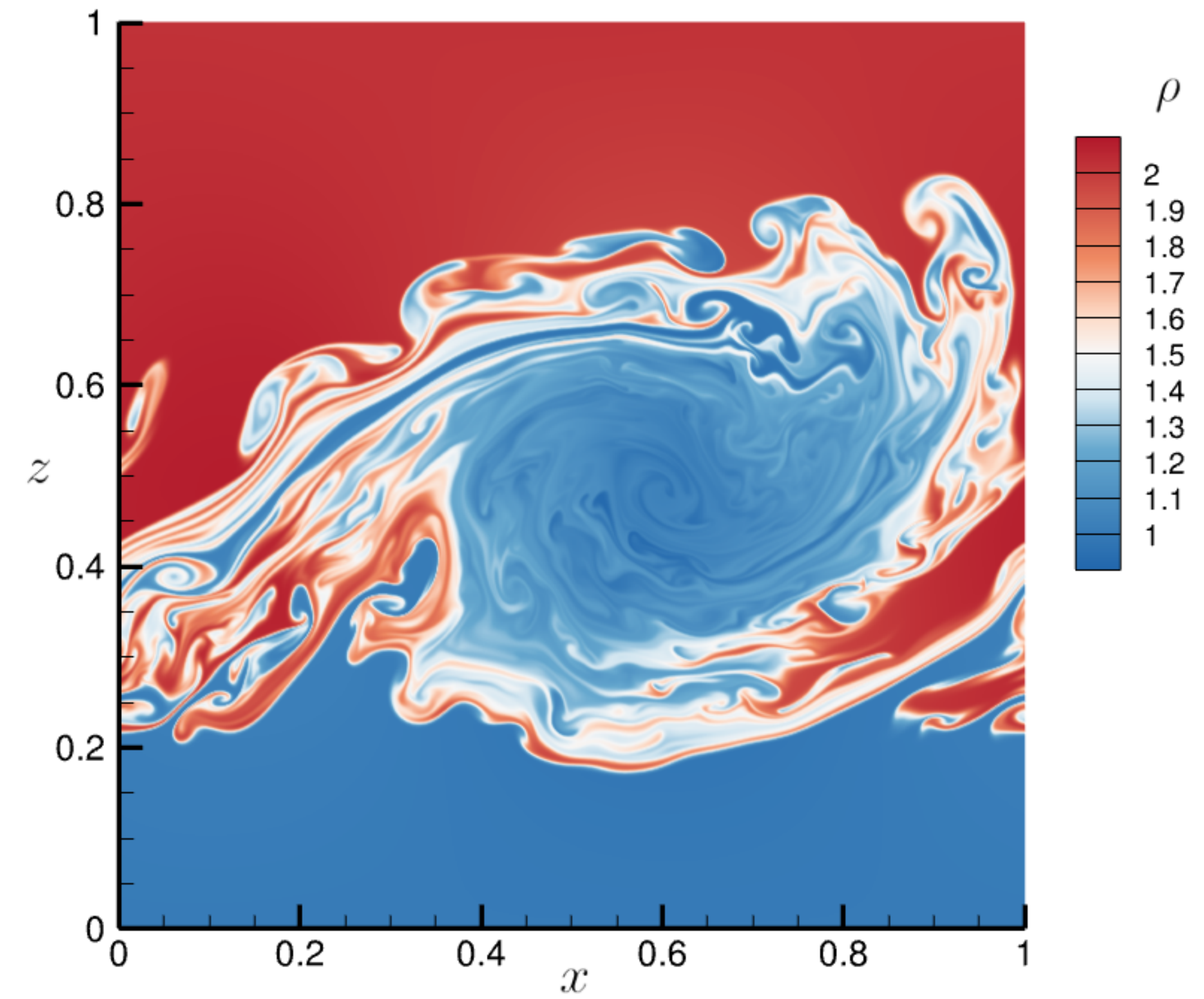
Figure from: Xie, J. et al. (2021). A low-communication-overhead parallel method for the 3D incompressible Navier-Stokes equations. [10.48550/arXiv.2104.08863](https://arxiv.org/abs/2104.08863).

Data from Daniel Lecoanet (Princeton)

(a) Padé code, $n_x = 1024$.



(b) Dedalus code, $n_x = 4096$.



VSI Simulations (locally isothermal vertically)

$$512(r) \times 512(z) \times 1024(\phi)$$

Same basic state disk parameters (temperature and density exponents) as Manger et al. (2020) : Pluto, PPM Scheme.

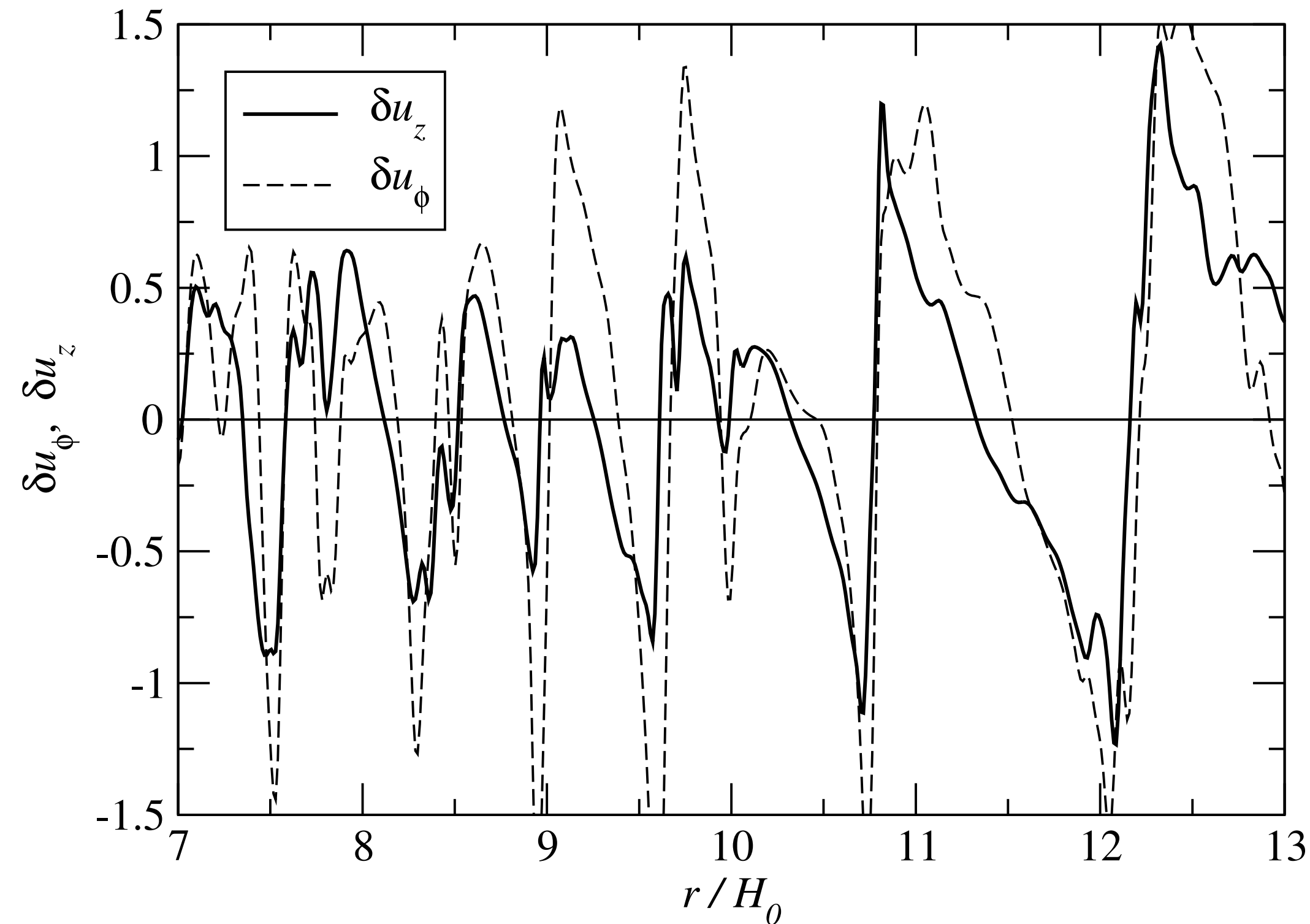
4.4 times the number of points per H_0 in r and z as Manger et al.
Similar number of points per H_0 in azimuth.

Our radial domain size is half theirs.

Lesur et al. (2025) beat everyone in resolution and radial domain size. Compact 3rd order shock-capturing scheme.

Axisymmetric Dynamics (present in 3D in weaker form)

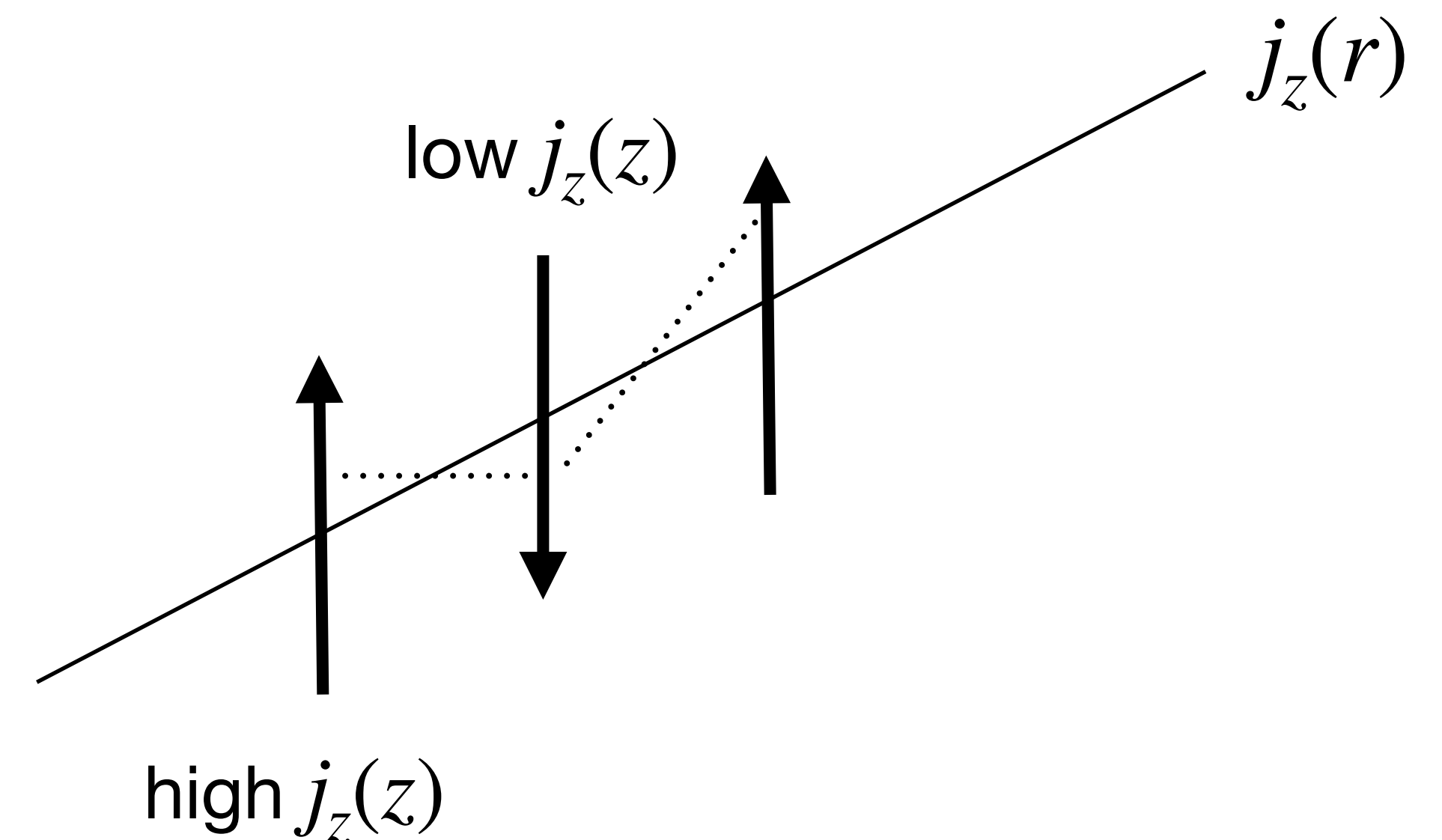
(a) N waves in perturbation velocities at $z = 2H_0$



Note: The profile, $u_z(r)$, of the vertical jets is also N-shaped $\iff \delta\omega_\phi \lesssim 0$ dominates for $z \gtrsim 0$ (mirrored N-waves for $u_z(r)$ in $z < 0$).

Due to transport (by the jets) of angular angular momentum in the presence of mean radial and vertical gradients of angular momentum

N-shaped profiles of $\delta u_\phi \iff \delta\omega_z > 0$ dominates $\iff \dot{j}_z$ staircase (Melon Fuksman et al.)



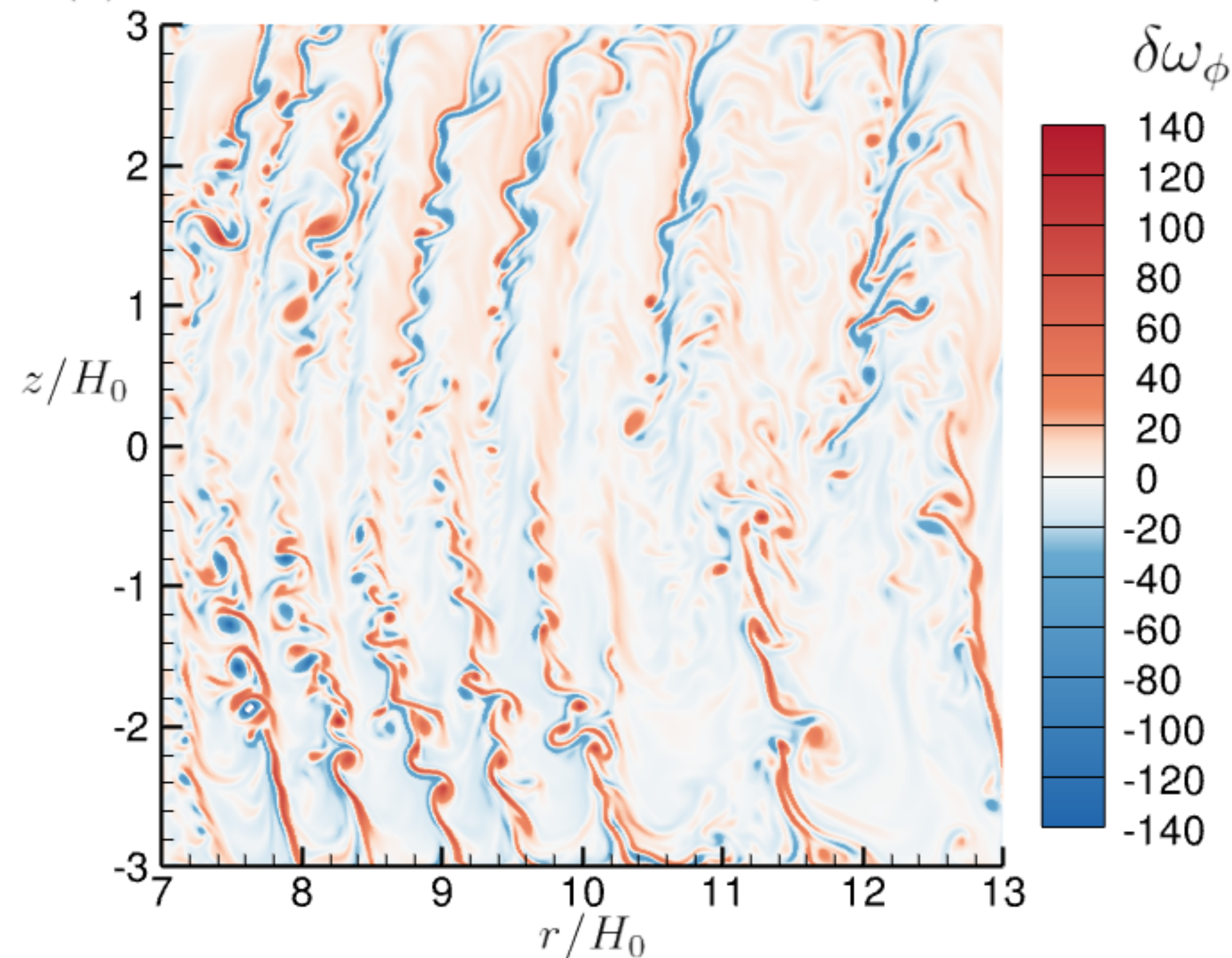
Axisymmetric Dynamics (cont.)

To explain sign pattern of ω_ϕ consider azimuthal vorticity equation (axisymmetric):

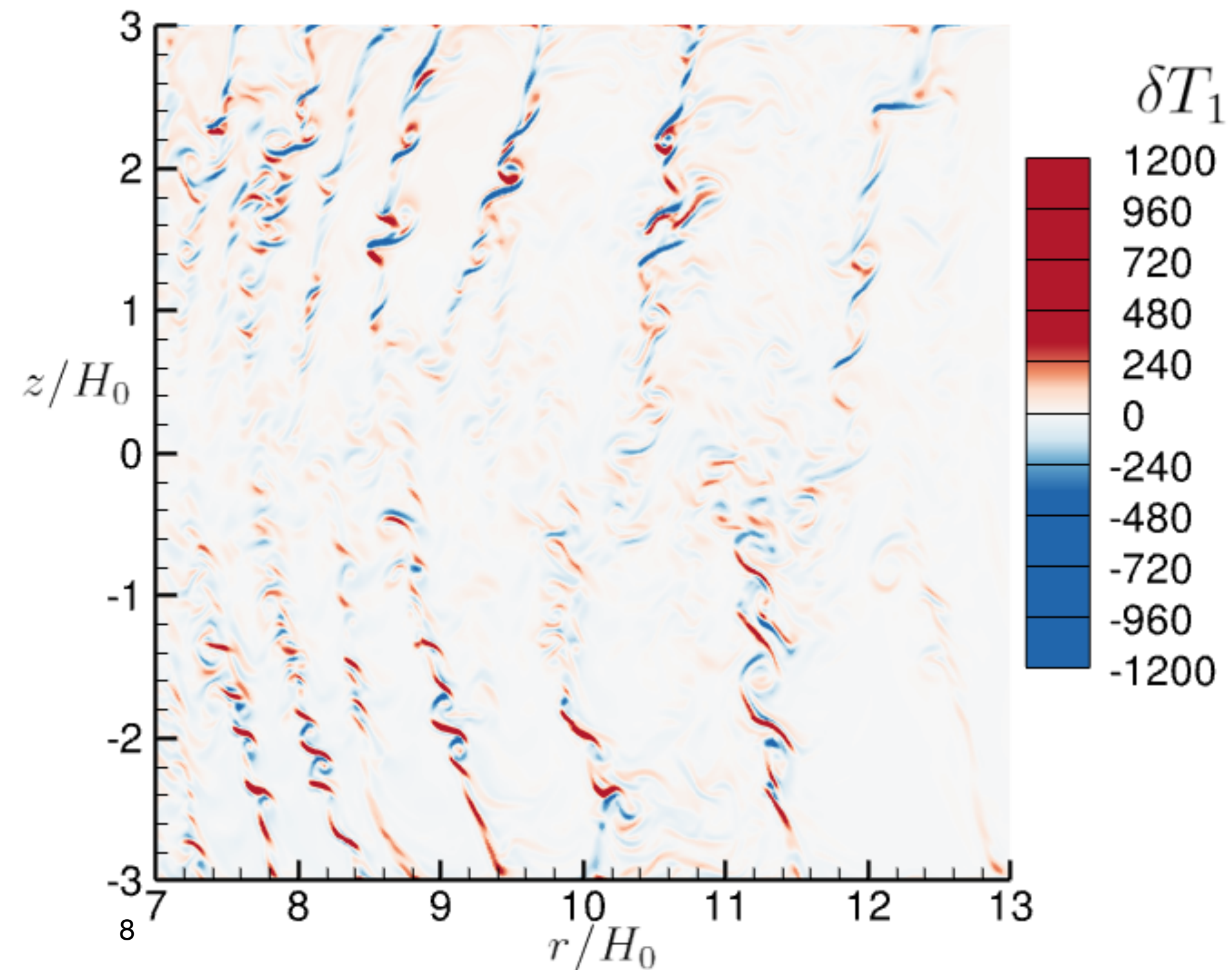
$$\underbrace{\frac{D}{Dt} \left(\frac{\omega_\phi}{\rho r} \right)}_{T_1} = \underbrace{\frac{1}{\rho r^2} \frac{\partial u_\phi^2}{\partial z}}_{T_2} + \frac{1}{\rho^2 r} \frac{\partial c_i^2}{\partial r} \frac{\partial \rho}{\partial z}.$$

$\delta T_1 \gg \delta T_2$ and much larger than basic state T_1 and has roughly the required sign.

(a) Perturbation azimuthal vorticity at $t/T_0 = 36.1$.



(a) δT_1 , $t/T_0 = 24.5$, Range = $[-1200, 1200]$.



Note: Melon-Fuksman et al. emphasize vertical advection terms. We emphasized source terms. Both should be considered.

Possible feedback loop if above picture is correct

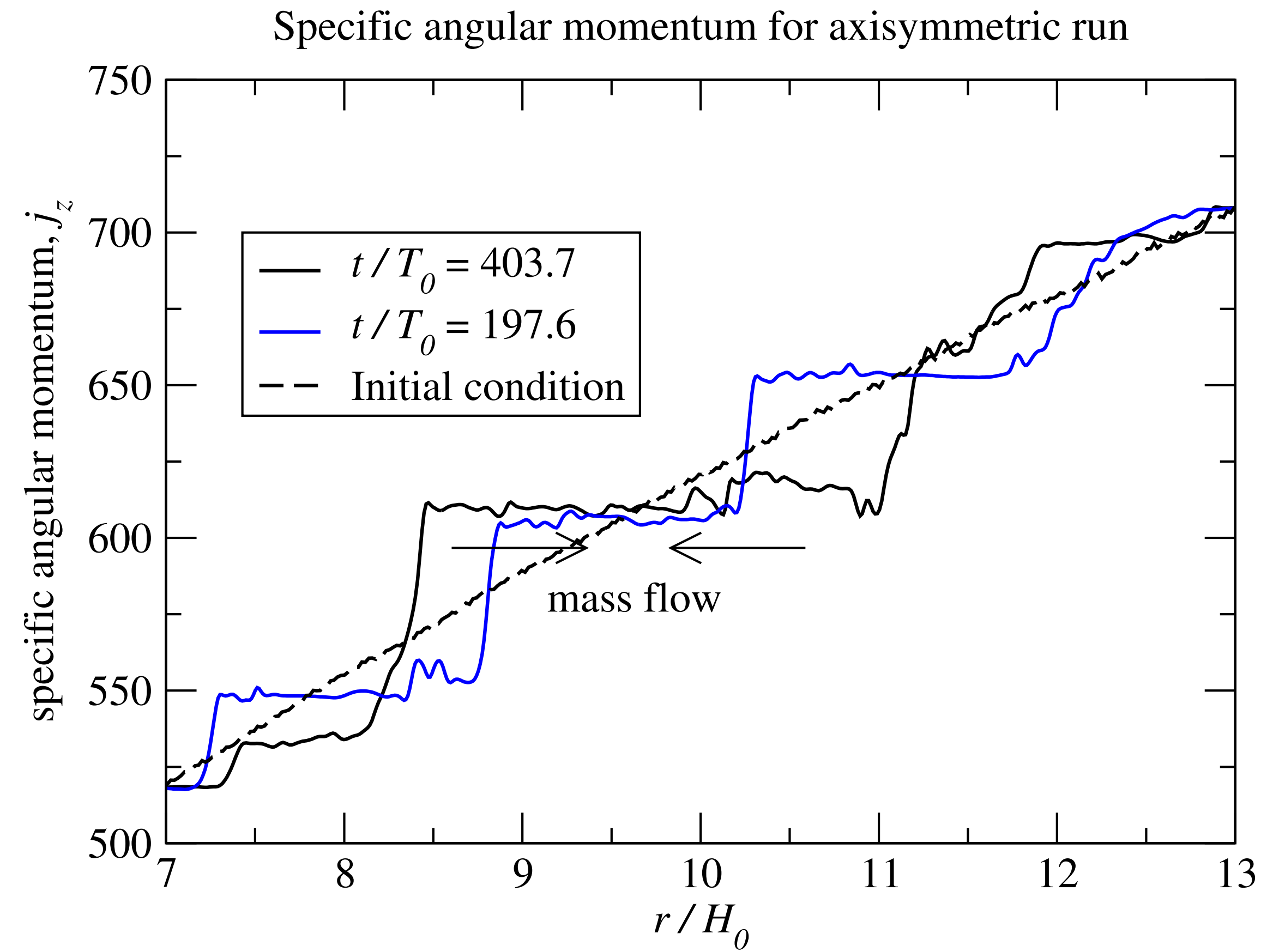
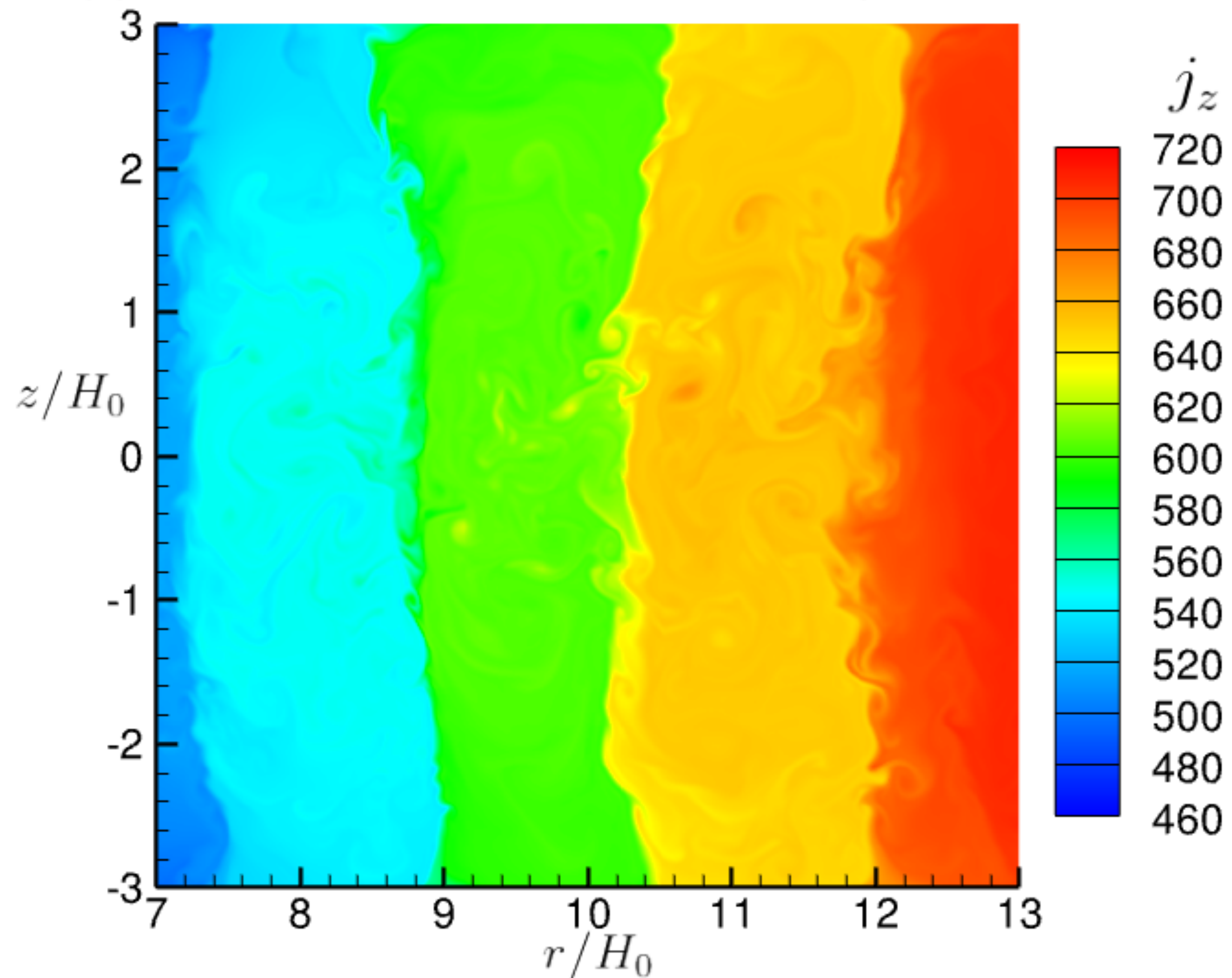
Vertical Jets (ω_ϕ) $\implies \delta u_\phi$ via vertical angular momentum transport by jets



Axisymmetric Dynamics (cont.)

Eventually the flat regions merge. No physical stationary state (confirmed in Fuksman et al. simulations with larger radial domain — false stationary states.)

(c) Specific angular momentum at $t/T_0 = 197.6$.

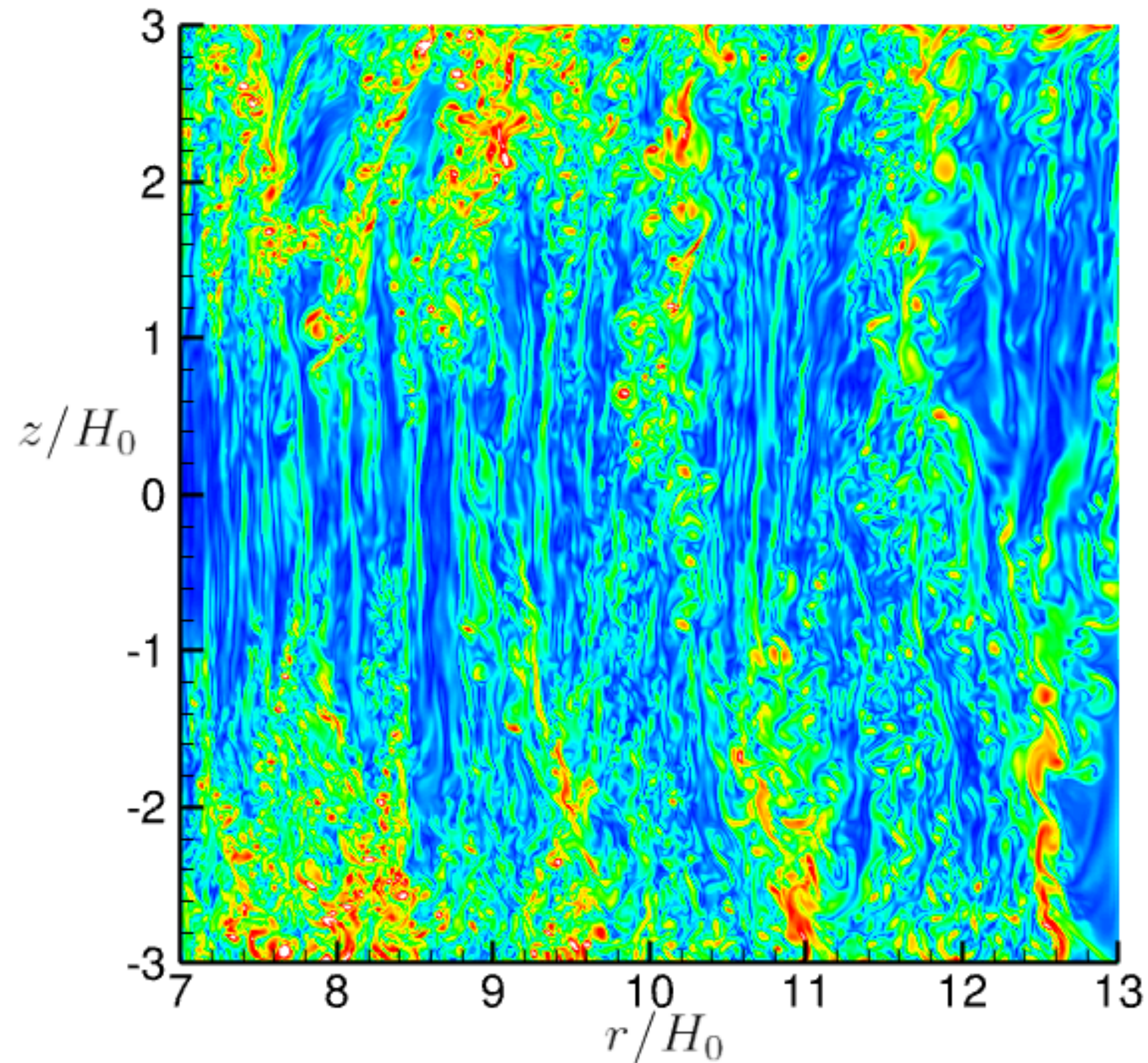


The strong staircase and merging are an artifact of axisymmetry.

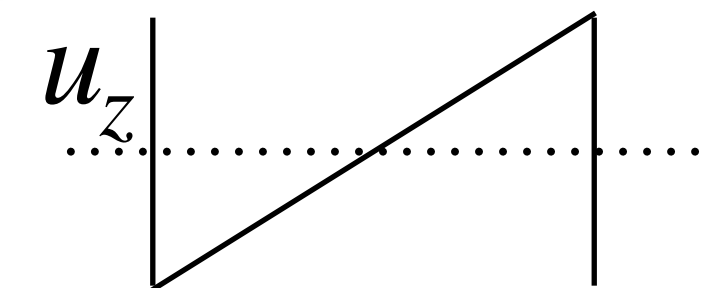
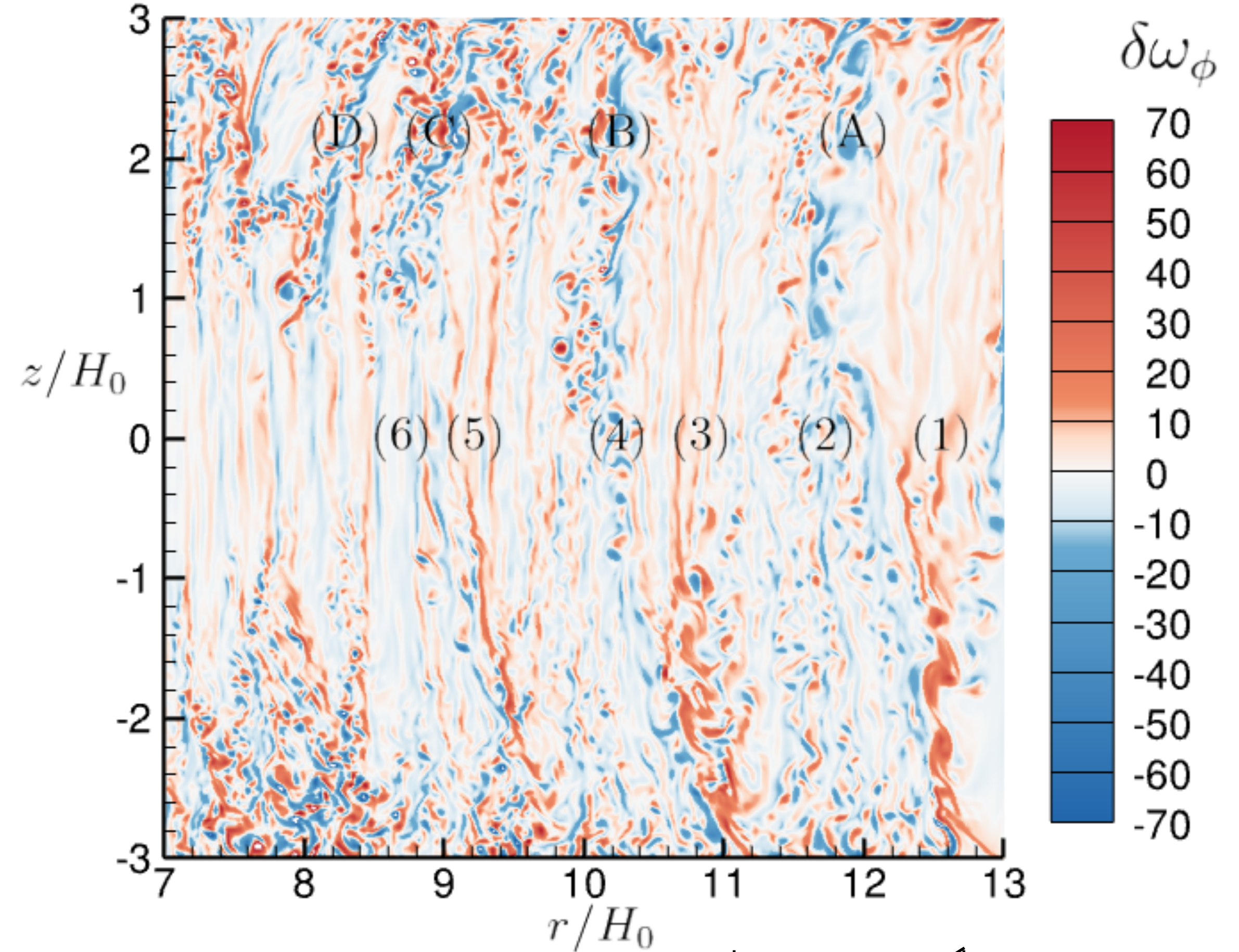
Now 3D

Preference for $\omega_\phi \lesssim 0$ for

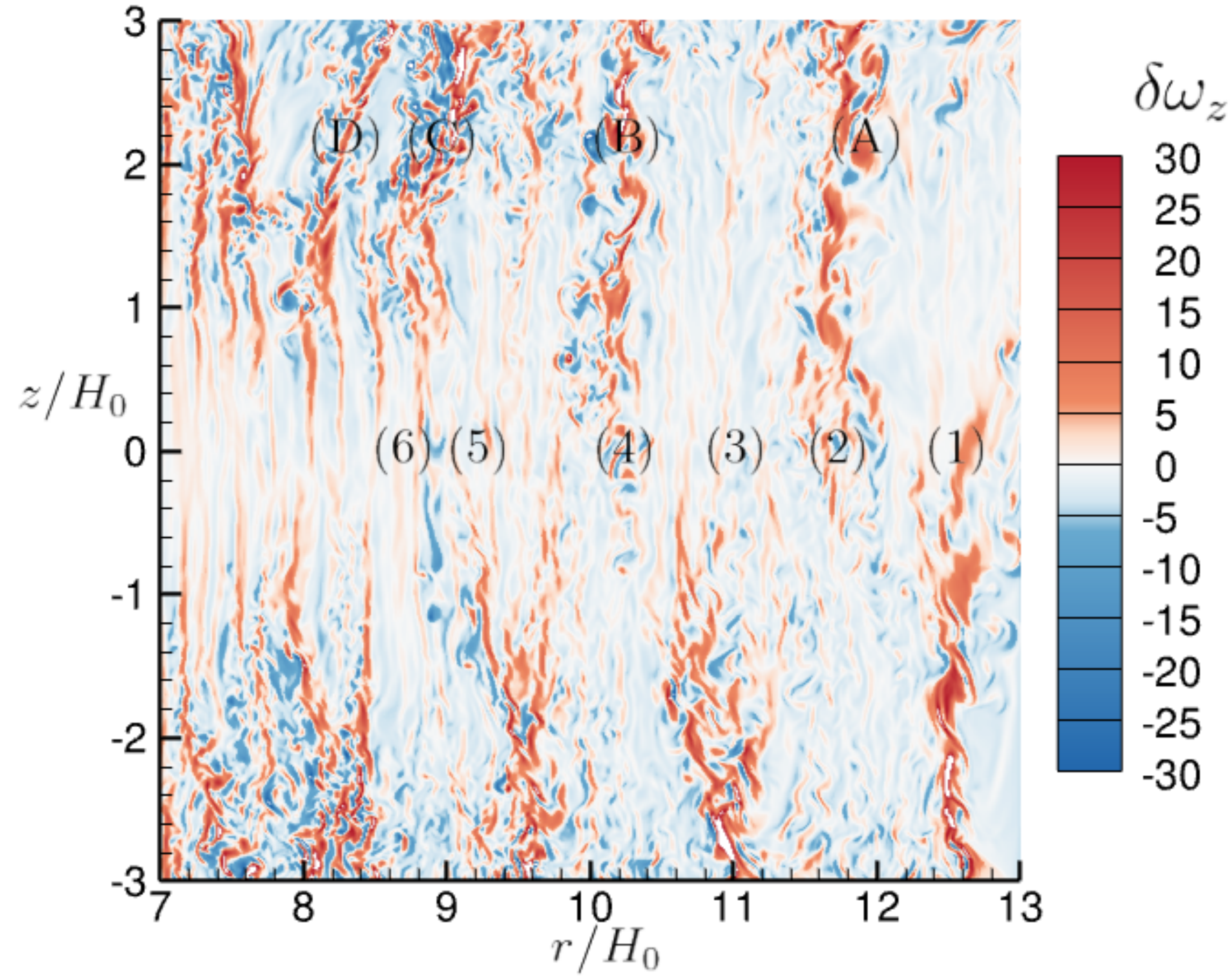
(a) Magnitude of perturbation vorticity, $t/T_0 = 444.64$.



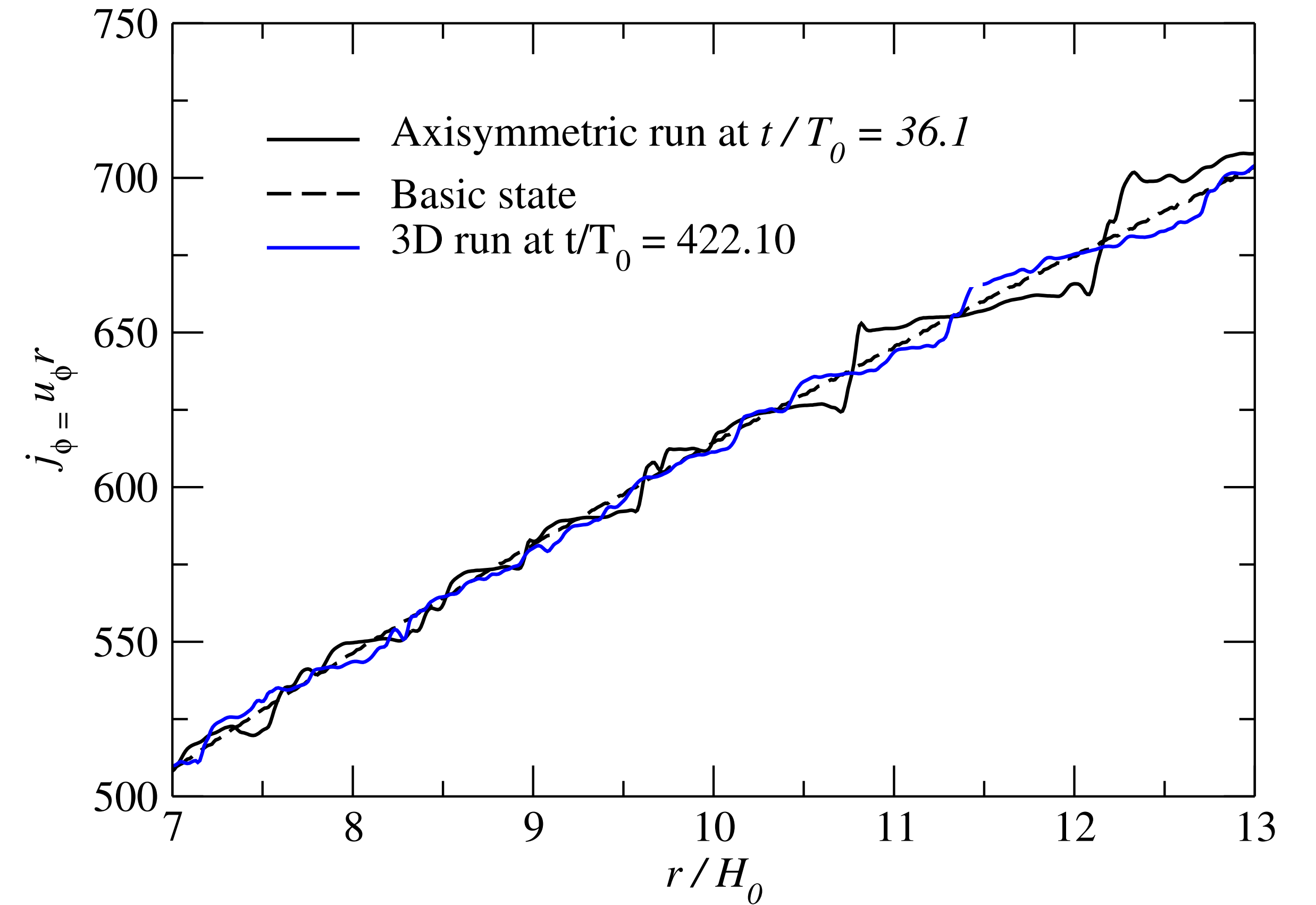
(b) Perturbation azimuthal vorticity, $t/T_0 = 444.64$.



(c) Perturbation vertical vorticity, $t/T_0 = 444.64$.



(b) Specific angular momentum at $z = 2H$.



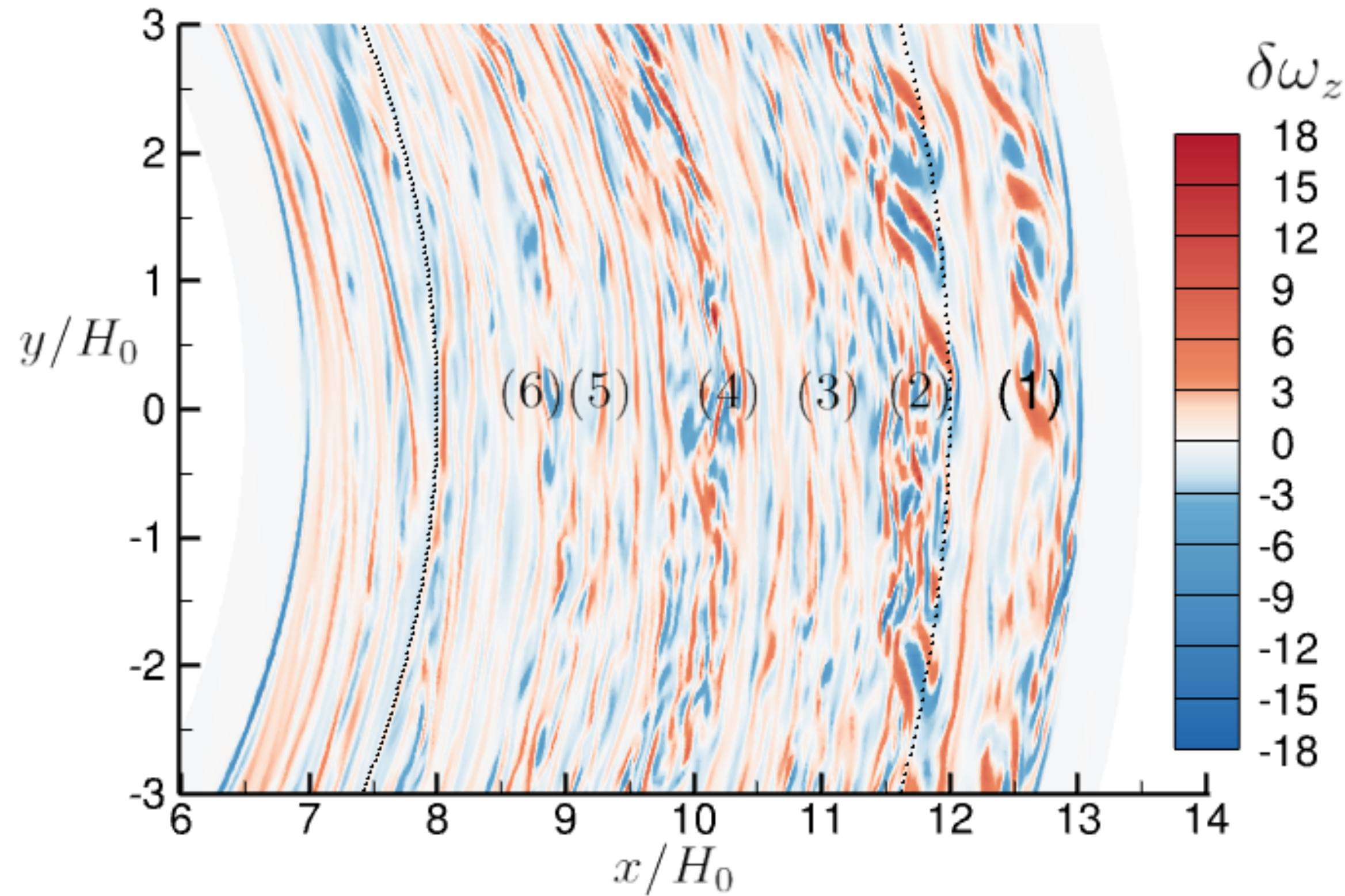
Preference for $\delta\omega_z > 0$ (and j_z staircase), but smaller than in axisymmetric case.

Perturbation vertical vorticity

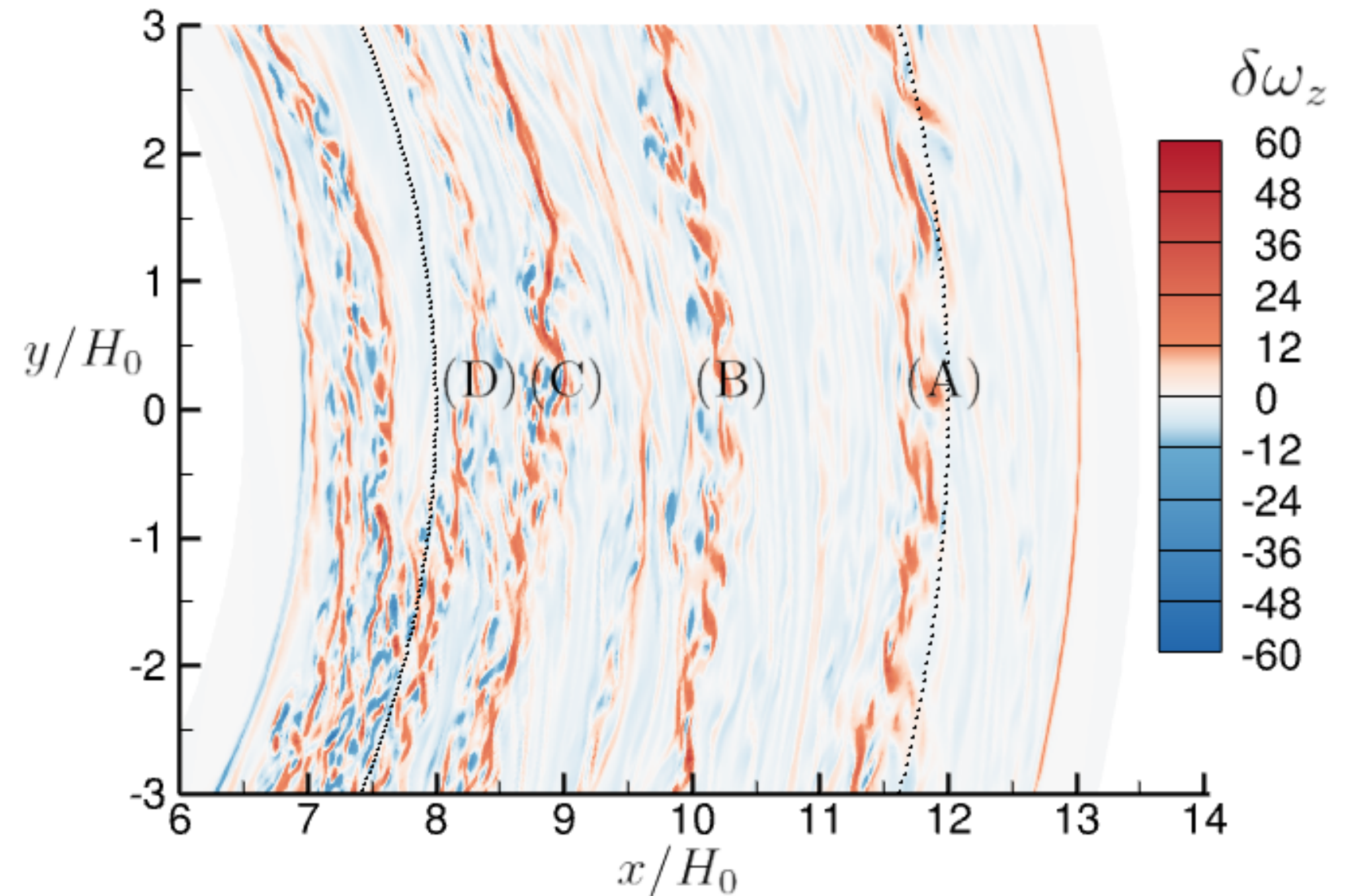
Midplane

$$z = 2H_0$$

(a) Perturbation vertical vorticity $z = 0, t/T_0 = 444.64$.

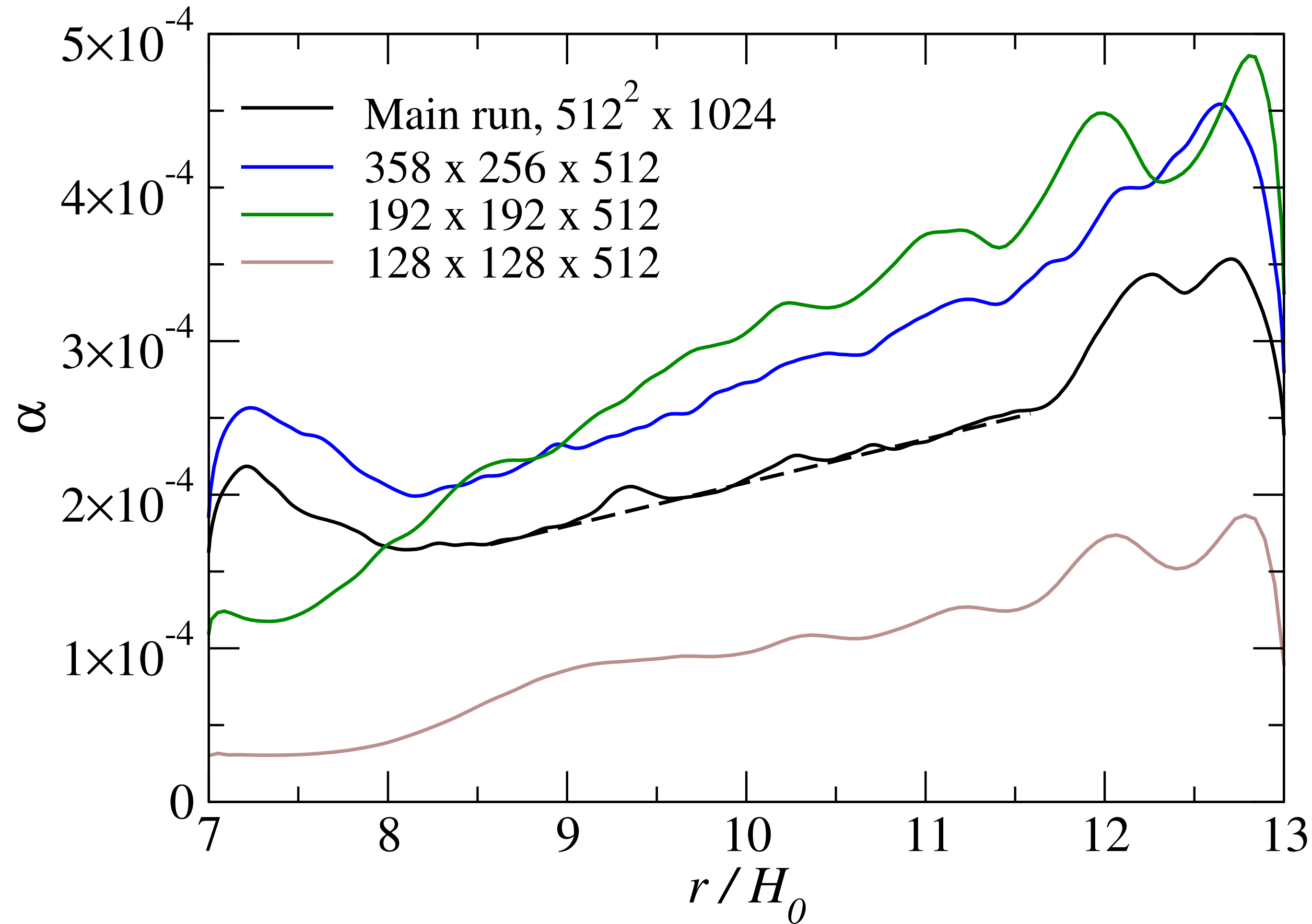


(a) $\delta\omega_z, z/H_0 = 2.0, t/T_0 = 444.64$.



No large scale persistent vortices (more detailed discussion in paper) reported by Manger et al. Confirmed by Lesur et al. (2025)

α parameter

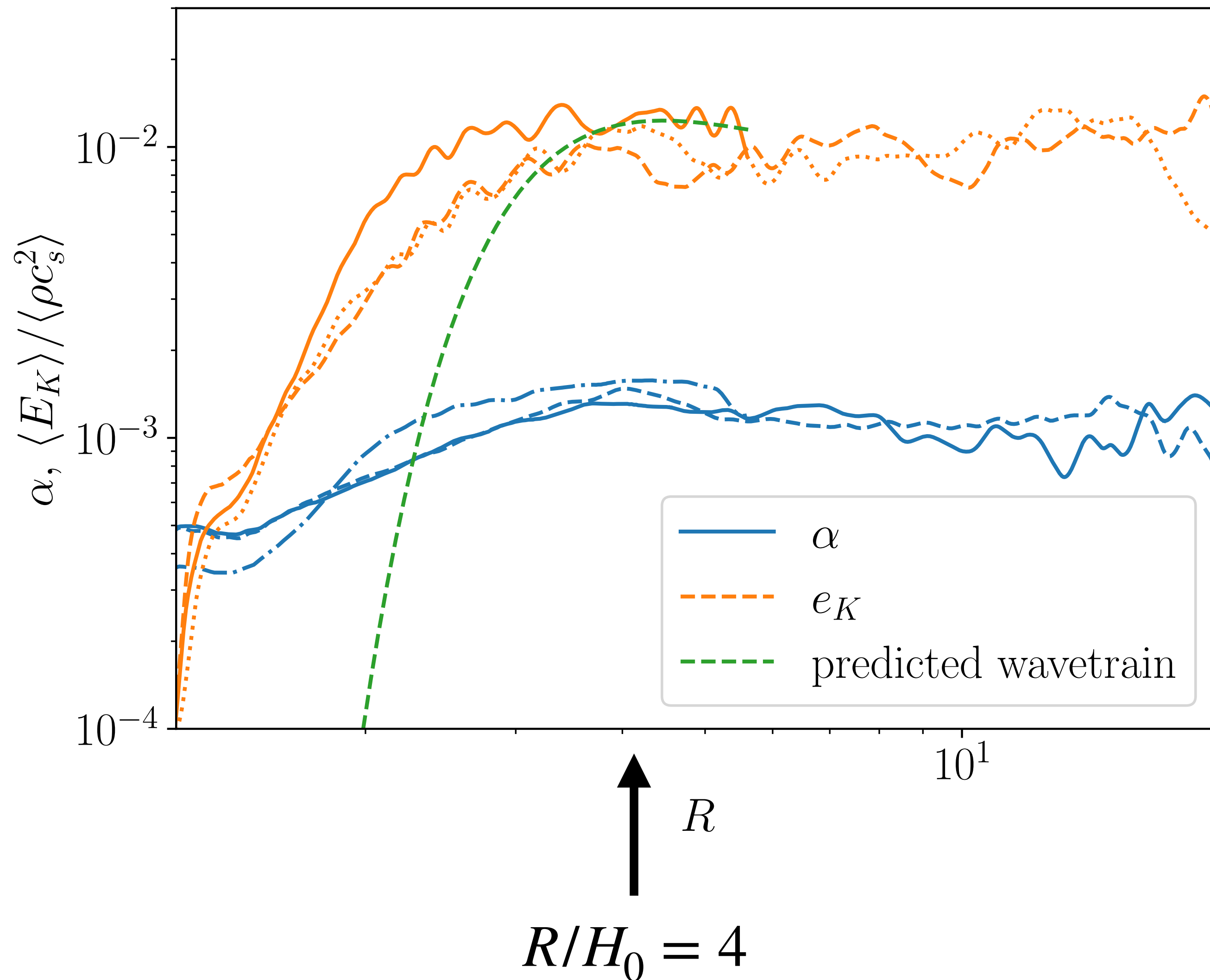


(1) Would like to see convergence with increasing resolution and radial domain size.

(2) The radial ends are contaminated by the wall boundary condition. The left peak in α leads to an artifactual RWI generated long-lived vortex.

(3) $\alpha(r)$ grows with r in the inner region. More on the next slide.

From Lesur etal. (2025)



(4) Lesur etal suggest that the increase of $\alpha(r)$ with r is an artifact of the left (wall) boundary which has an effect deep into the disk via a radially propagating inertial wave.

(5) We agree, or at least, take their suggestion very seriously.

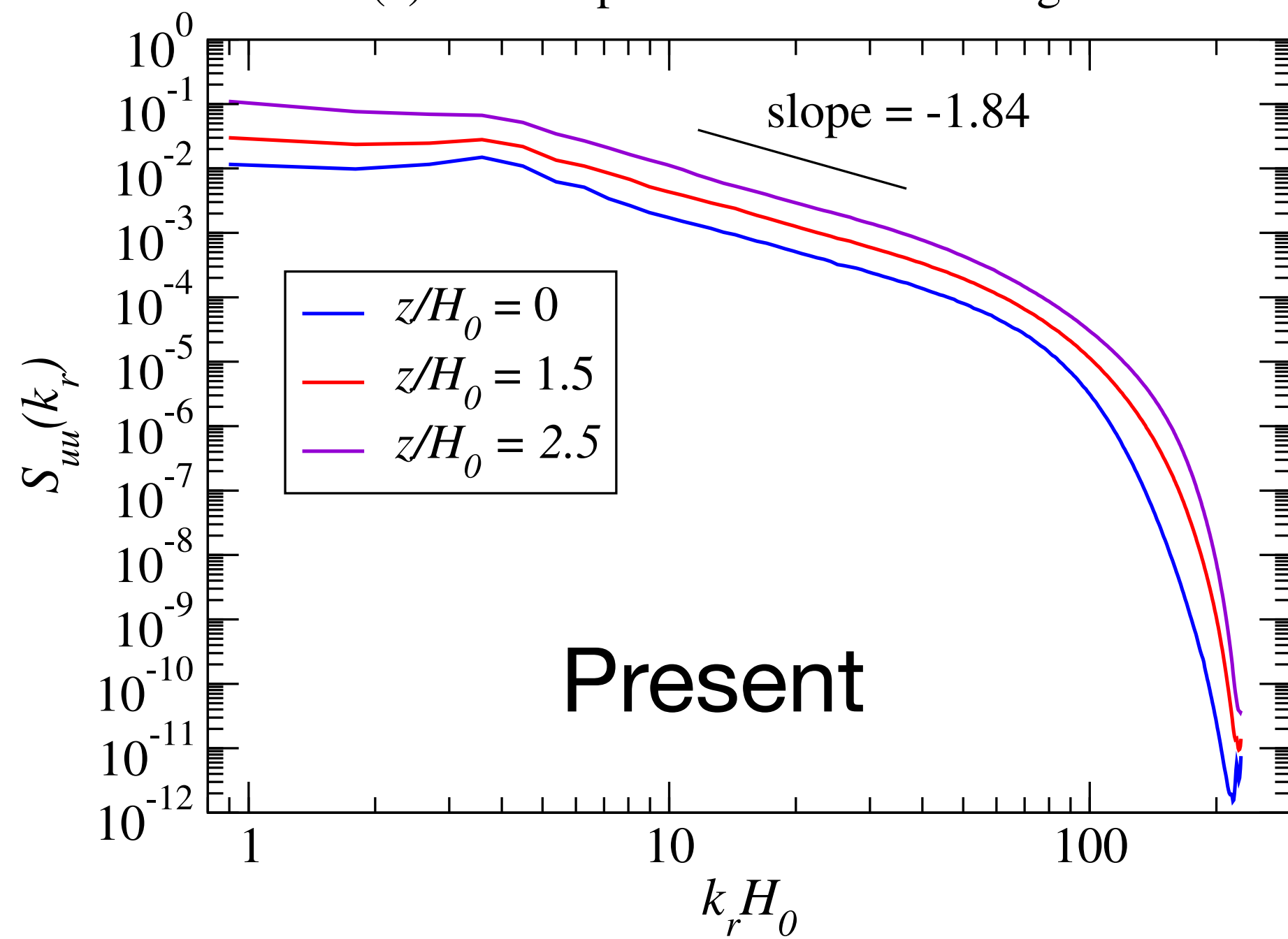
(6) Our value of α is also smaller compared to their value and that of Manger etal (2020), perhaps due to similar reasons.

Finally, Spectra

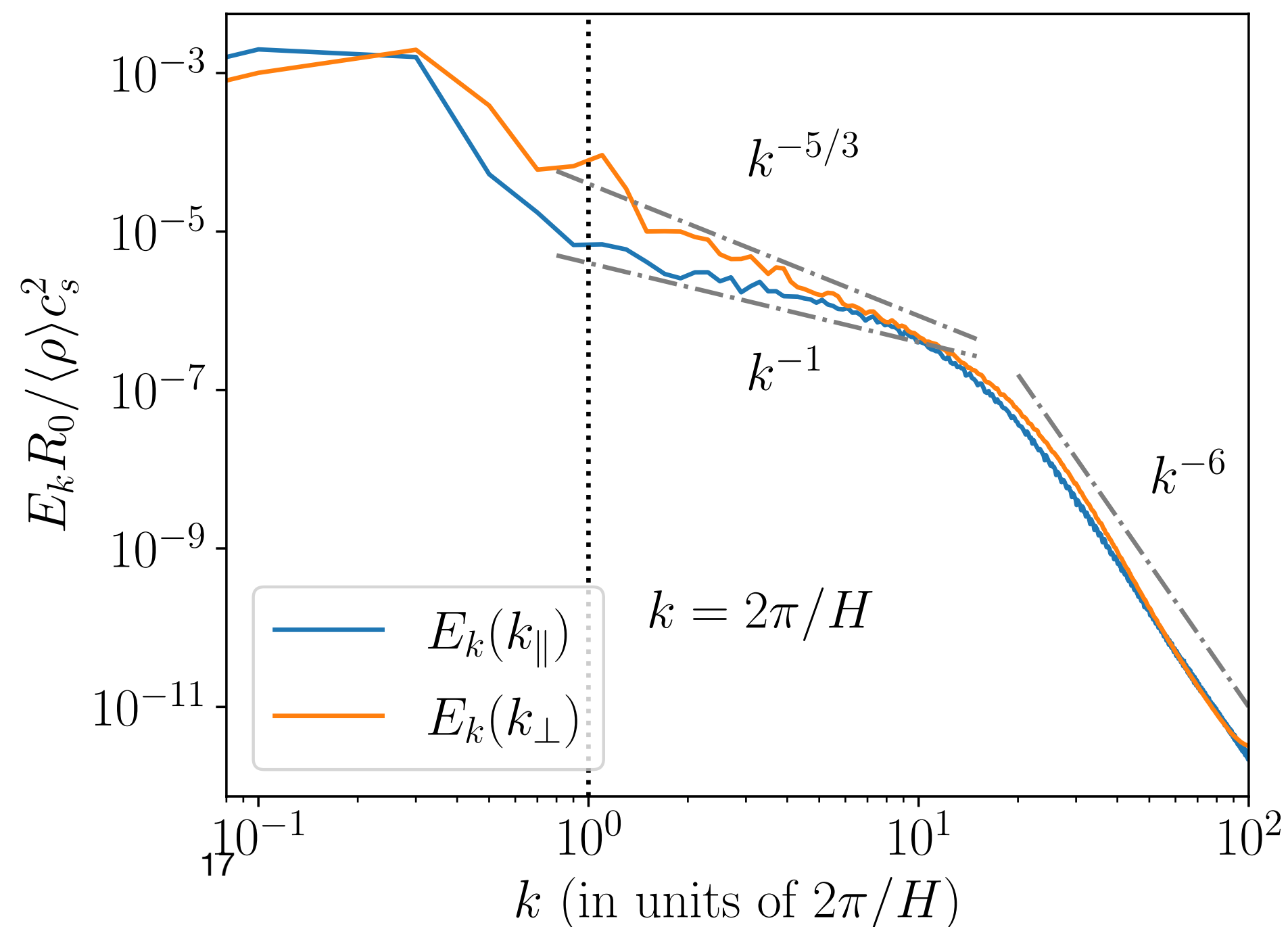
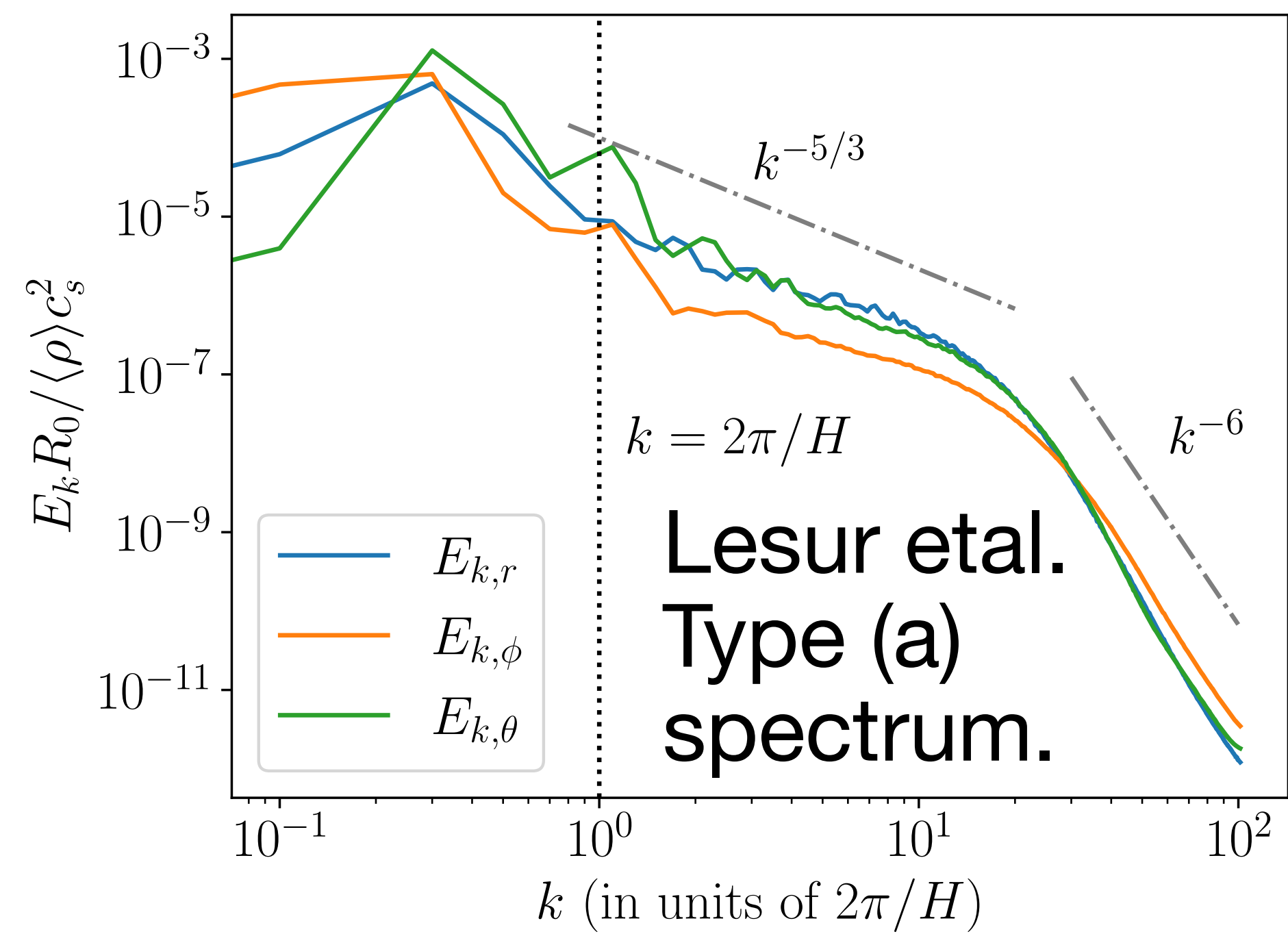
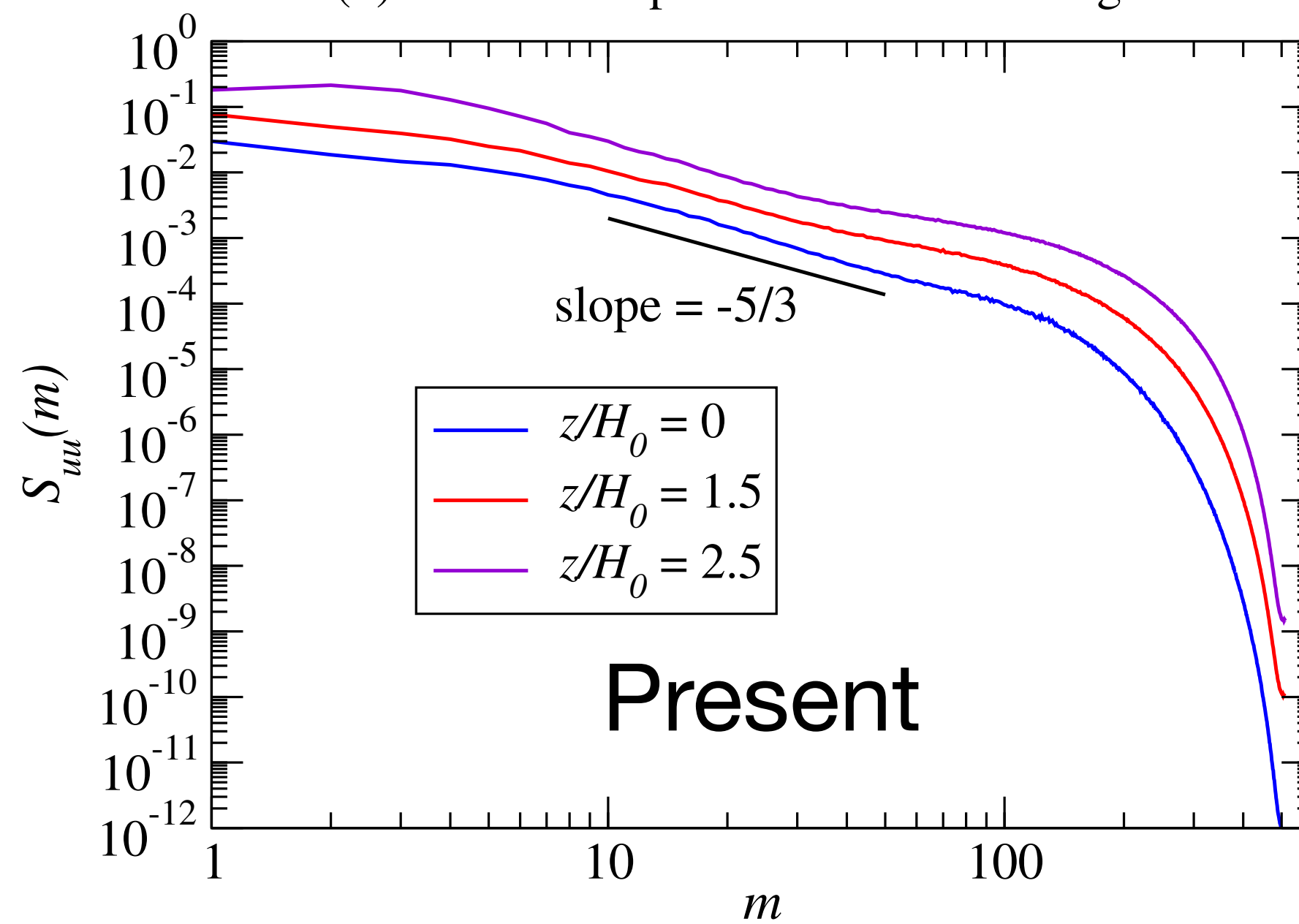
Procedural Issues

- (1) Since the only (statistically) homogeneous direction is ϕ , using Fourier, the only rigorous velocity power spectrum is $E(R_0, z_0, k_\phi)$, in cylindrical coordinates, say. But this is too restrictive.
- (2) Our current philosophy: consider z_0 's separately, but allow spectra wrt k_R (with Hanning window) as well as averaging of k_ϕ spectra wrt R .
- (3) Lesur et al. (2025): (a) Cubic box, roughly $4H_0$ to a side, centered at fixed r_0 and $\theta = \pi/2$, Hanning window, and spectra wrt spherical shells in $\mathbf{k}$ space. (b) In the same box, spectra wrt $k_\perp = k_r$ and $k_\parallel = (k_\phi^2 + k_\theta^2)^{1/2}$. BTW, why not $k_\parallel = k_z$ and $k_\perp = (k_x^2 + k_y^2)^{1/2} = k_R$?
- (4) Workers in the field should decide on some standard procedures to allow intercomparisons and learning physics. Not currently the case.

(a) Radial spectrum at various heights.



(b) Azimuthal spectrum at various heights.

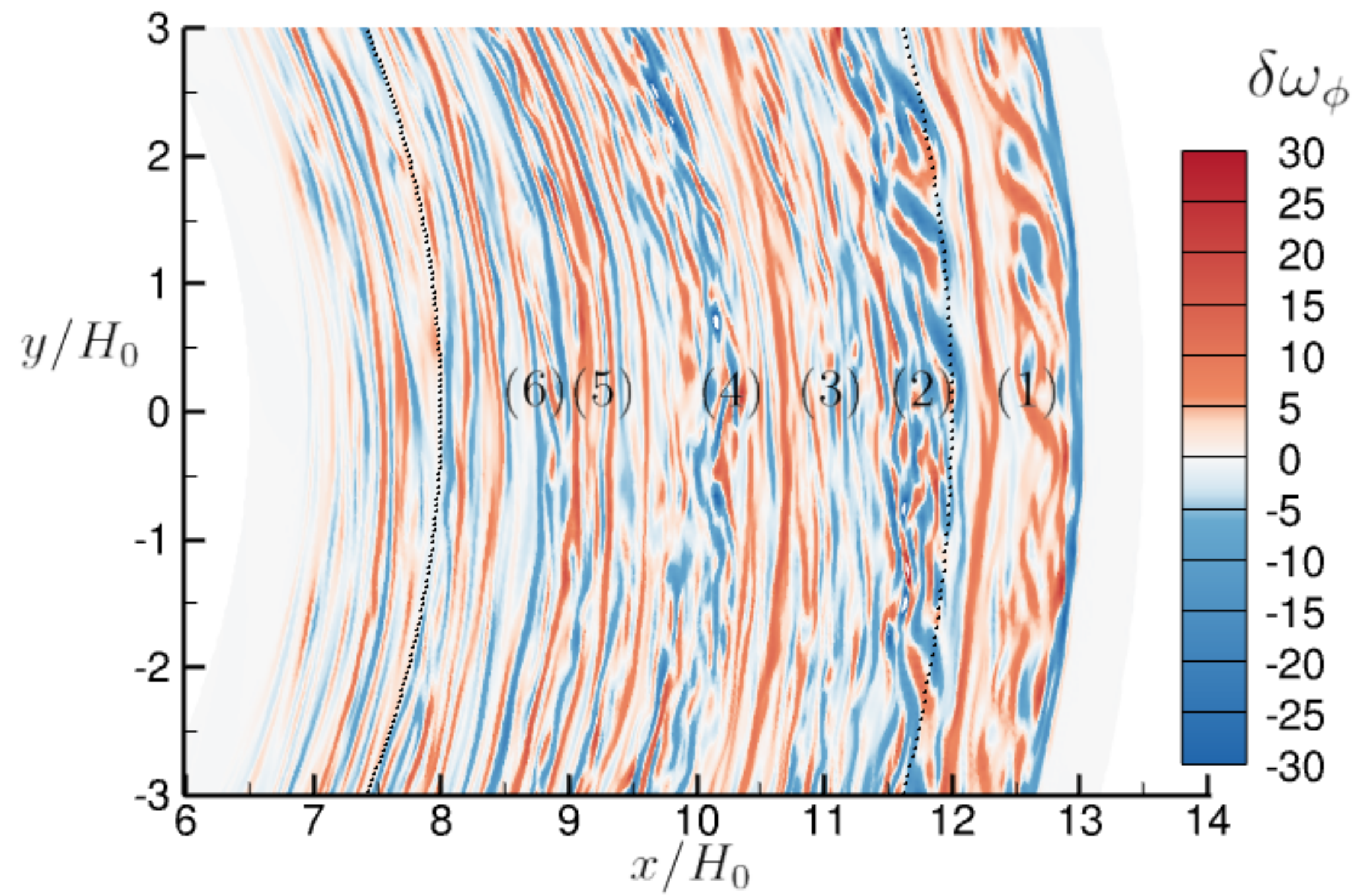


“This suggests that the break length scale follows the resolution and hence that the k^{-6} part of the spectrum is due to numerical diffusion.” — Lesur et al. (2025)

Important question: Does the non-linear dissipation of shock-capturing methods produce a k^{-6} dissipation for low Mach number vortical turbulence? Note: $k^{-4} \implies$ sharp vorticity jumps across vortex patches (P. Saffman).
 $k^{-6} \implies ???$

Thank-you for your attention.

(b) Perturbation azimuthal vorticity $z = 0, t/T_0 = 444.64$.



(b) $\delta\omega_\phi$, $z/H_0 = 2.0, t/T_0 = 444.64$.

